

Data Quality Report — Q2 (Real Data)

One page. Written form of the Q2 answer on the **real** 10,547-review corpus. Full JSON: data/processed/defect_detection_report_real.json. Run standalone: python3 src/real/detect_defects_real.py.

Product-inclusion quality (a real-data problem a synthetic fixture cannot have)

Detection. "Appliances" metadata yielded only 32 candidates, dominated by replacement-filter accessories misclassified as purifiers. Switched to "Home_and_Kitchen"; a keyword classifier there produced real false positives — vacuum cleaners, wearable necklace ionizers, home-decor products whose *description* mentioned "air purifier" in cross-sell copy. **Remedy.** Tightened to title-only matching with regex exclusions, then a minimum 5-real-rating threshold. The verifiable chain, each step a committed file: 32 Appliances candidates + the reclassified Home_and_Kitchen pool → 309 merged candidates (purifier_products_final.jsonl) → **237 final** (purifier_products_frozen.jsonl), re-inspected by hand at each step. **Effect of not catching it.** The corpus would have included hundreds of replacement-filter and vacuum-cleaner reviews under an "air purifier" label, corrupting every downstream prevalence and CSAT figure.

Defect: rating/text sentiment contradictions

Detection. Negation-aware polarity scoring (rating==5 with strongly negative text, or rating==1 with strongly positive text). **Detected: 239 / 10,547 (2.27%).** **Remedy.** Quarantined (rating_trusted=false), not deleted — the text is real experience, only the star label is untrusted. **Quantified same-corpus consequence (computed, not asserted — see quantified_consequence in the JSON report).** The corpus mean star rating — the baseline every Q3 CSAT Impact is measured against — is **4.148★ without the remedy** (all rows) and **4.172★ with it** (trusted rows only): an absolute shift of **+0.024★ (+0.59%)**, driven entirely by the 239 quarantined rows (their own mean is 3.09★). Deliberately small and reported at its true size — but it moves the baseline under every per-theme CSAT figure in Q3/Q6. **Detector-development note.** The first detector version had no negation handling and misread "I cannot recommend this product" as *positive*; an 18-character negation window plus a real-vocabulary term list landed at the final 239. Useful debugging history — kept separate from the same-corpus consequence above, which is the number that answers Q2.

Defect: empty/trivial review text

Detection. len(review_text.strip()) < 3 characters. **Detected: 18 / 10,547 (0.17%).** **Remedy.** Removed — no analyzable signal.

Non-defects (real, honest findings)

0 exact-duplicate-text bursts (≥3× on the same product) and **0 per-product daily volume anomalies** (robust z-score ≥5). A genuine result, not a detector failure — absence of a defect is itself a finding worth stating plainly.

Summary

Defect	Count / Corpus	Remedy
Product misclassification (pre-corpus)	309 merged candidates → 237 final (72 dropped by threshold + exclusions)	classifier tightened, re-inspected by hand
Sentiment/rating conflicts	239 / 10,547 (2.27%)	quarantined (rating_trusted=false)
Empty/trivial text	18 / 10,547 (0.17%)	removed
Duplicate-text bursts	0 (real finding)	none needed
Volume anomalies	0 (real finding)	none needed